Practice Quiz: Metallurgical Systems — Module 08: Planning

Name: ___ ***Date***: ___

Part A: Multiple Choice

1. In the Active Inference framework, planning in a metallurgical system is best understood as:

A) A random process with no thermodynamic driving force B) A process driven by the minimization of free energy (both thermodynamic and variational) C) An externally imposed constraint with no internal dynamics D) A purely kinetic phenomenon unrelated to equilibrium

1. The Markov blanket of a crystal grain includes:

A) Only the atoms at the grain interior B) The grain boundary — mediating interactions between the grain's internal states and external environment C) The entire polycrystalline sample D) Only the external temperature field

1. In the context of planning, "prediction error" in a metallurgical system corresponds to:

A) A measurement instrument malfunction B) The deviation between the system's current state and its equilibrium (predicted) state C) An error in the periodic table D) The difference between two competing phase diagrams

1. Which characterization technique provides the highest precision for studying crystallographic planning?

A) Visual inspection B) Hardness testing C) Electron backscatter diffraction (EBSD) D) Density measurement

1. The thermodynamic free energy G = H - TS and the variational free energy F share:

A) No mathematical relationship B) The same fundamental structure — both measure departure from equilibrium C) Identical numerical values in all cases D) Only a metaphorical connection

1. When a supersaturated Al-Cu alloy forms GP zones during aging, this is an example of:

A) The system increasing its free energy B) Active inference — the system acting to minimize its thermodynamic free energy C) A violation of the Second Law D) Random atomic motion with no driving force

1. Nested Markov blankets in metallurgy refer to:

A) Multiple layers of insulation around a furnace B) The hierarchical structure: atoms -> unit cells -> grains -> phases -> components C) A type of heat treatment schedule D) Overlapping X-ray diffraction patterns

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Part B: Short Analysis

1. A martensitic transformation in steel occurs in milliseconds, while pearlitic transformation takes minutes to hours. Using Active Inference language, explain why these two "actions" of the system operate at such different timescales. What does this tell us about the system's inference process?

2. You have two characterization methods for studying planning: optical microscopy (fast, low cost, limited resolution) and TEM (slow, expensive, atomic resolution). Using the concept of epistemic value and precision, explain how you would decide which to use for a given research question.

3. Describe how a CALPHAD database functions as a "generative model" for a metallurgical system. What happens when the database predictions diverge from experimental observations? How is the model updated, and what is the Active Inference term for this process?